Pre- Requirement Software and Python library

1. Software

- Python 3.8 above

2. Python library

- Ultralytics
- Opencv-python
- Numpy

Steps to install,

pip install ultralytics opencv-python numpy

YOLOv3 Model Training

Dataset Configuration (data.yaml)

What This Does

This file tells YOLO:

- Where the images are
- What object classes exist

Step

1. Open **01_data.yaml** file.
2. Change the class and path according to the dataset that you want to train.

Example

```yaml
# ------------------------------------------------
# Dataset paths
# ------------------------------------------------

# Path to training images
train: ./01_Train_images/train/images

# Path to validation images
val: ./01_Train_images/valid/images

# Path to test images (optional)
# Uncomment if you have a test set
test: ./01_Train_images/test/images


# ------------------------------------------------
# Class names
# ------------------------------------------------

# List of object classes in the dataset
# Index starts from 0
names:
  - birds

```

- **Train** =images used to train the model
- **Val**=images used to check training performance
- **Names** = list of object classes (only one class: birds)

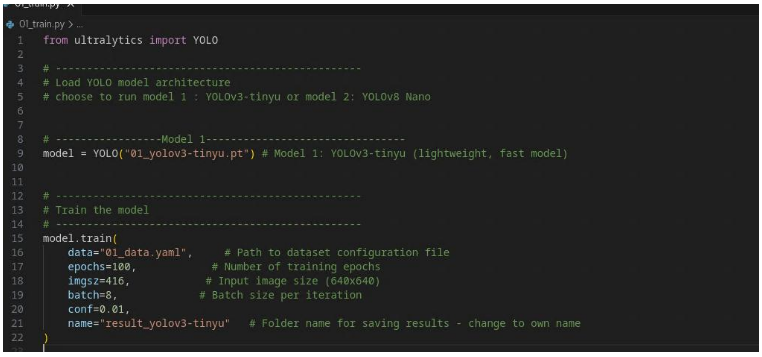
Train The YOLO Model

What This Does

- Loads a YOLO model
- Uses the processed images
- Trains the model to detect object

Step

1. Open **01_train.py**, make sure is 01_yolov3-tinyu.pt model is selected



```
01_train.py > ...
from ultralytics import YOLO

# ------------------------------------------------
# Load YOLO model architecture
# choose to run model 1 : YOLOv3-tinyu or model 2: YOLOv8 Nano


# ----------------Model 1------------------------------
model = YOLO("01_yolov3-tinyu.pt") # Model 1: YOLOv3-tinyu (lightweight, fast model)


# ------------------------------------------------
# Train the model
# ------------------------------------------------
model.train(
    data="01_data.yaml",     # Path to dataset configuration file
    epochs=100,             # Number of training epochs
    imgsz=416,             # Input image size (640x640)
    batch=8,              # Batch size per iteration
    conf=0.01,
    name="result_yolov3-tinyu"   # Folder name for saving results - change to own name
)
```

2. Open the console and run the code
   python 01_train.py
3. To run another model, make sure the **runs** folder for the result is deleted first.

Results

After training, results are saved automatically in **runs** folder.

The folder contains:

- Trained model files are inside ./runs/detect/result_yolov3-tinyu/weights path: **best.pt**, **last.pt.** This model can be trained again by replacing 01_yolov3-tinyu.pt in step 1.
- Training logs
- Performance graphs

YOLO Model Inference With Custom Image

What This Does

- Loads a YOLO trained model
- Uses custom image
- Run the Model

Step

1. Copy the **best.pt** model from the train file : ./runs/detect/train/weights. Paste to the same directory as **02_Run_YOLO_Model.py.**
2. Open **02_Run_YOLO_Model.py** file
3. Replace input image directory to your own image location
4. Replace “name” for result to your own file name

```
from ultralytics import YOLO

model = YOLO("best.pt")   # get the best.pt from your  train result
results = model("./infr_images_s/", #image file directory #input image
                save=True, project="Inference", name="detection_result") #output image

print("Inference completed")
```

5. Run the code *python 02_Run_YOLO_Model.py* in console.

Results

After completed, results are saved automatically
in : ./runs/detect/Inference/detection_result for example.